

UNIT-2



HTML5 Basics – Formatting – Colors –
Images – Links – Tables – Lists – Layout–
Forms–Canvas–Media.

HTML

- HTML – **Hyper Text Markup Language** – The Language of Web Pages on the World Wide Web.

HTML is a text formatting language.

- HTML describes the structure of a Web page.
- HTML elements label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.
- `<!-- This is a comment -->`

Creating a Basic Starting Document

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>My First Heading</h1>
<h4>My First Heading</h4>
<p>My first paragraph.</p>

</body>
</html>
```

My First Heading

My First Heading

My first paragraph.

Creating a Basic Starting Document

- The `<!DOCTYPE html>` declaration defines that this document is an HTML5 document
- The `<html>` element is the root element of an HTML page
- The `<head>` element contains meta information about the HTML page
- The `<title>` element specifies a title for the HTML page (which is shown in the browser's title bar or in the page's tab)
- The `<body>` element defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.
- The `<h1>` element defines a large heading
- The `<p>` element defines a paragraph

HTML Formatting Element

- - Bold text
- - Important text
- <i> - Italic text
- - Emphasized text
- <mark> - Marked text
- <small> - Smaller text
- - Deleted text
- <ins> - Inserted text
- <sub> - Subscript text
- <sup> - Superscript text

Color Codes

- Colors are set using “**RGB**” color codes, which are, represented as hexadecimal values. Each 2-digit section of the code represents the amount, in sequence, of **red**, **green** or **blue** that forms the color. For example, a **RGB** value with 00 as the first two digits has no red in the color.

16 Basic Colors

Color Name	RGB Triplet	Hexadecimal	Color Name	RGB Triplet	Hexadecimal
Aqua	(0,255,255)	00FFFF	Navy	(0,0,128)	000080
Black	(0,0,0)	000000	Olive	(128,128,0)	808000
Blue	(0,0,255)	0000FF	Purple	(128,0,128)	800080
Fuchsia	(255,0,255)	FF00FF	Red	(255,0,0)	FF0000
Gray	(128,128,128)	808080	Silver	(192,192,192)	C0C0C0
Green	(0,128,0)	008000	Teal	(0,128,128)	008080
Lime	(0,255,0)	00FF00	White	(255,255,255)	FFFFFF
Maroon	(128,0,0)	800000	Yellow	(255,255,0)	FFFF00

Color Codes

1. WHITE
2. BLACK
3. RED
4. GREEN
5. BLUE
6. MAGENTA
7. CYAN
8. YELLOW
9. AQUAMARINE
10. BAKER'S CHOCOLATE
11. VIOLET
12. BRASS
13. COPPER
14. PINK
15. ORANGE

1. #FFFFFF
2. #000000
3. #FF0000
4. #00FF00
5. #0000FF
6. #FF00FF
7. #00FFFF
8. #FFFF00
9. #70DB93
10. #5C3317
11. #9F5F9F
12. #B5A642
13. #B87333
14. #FF6EC7
15. #FF7F00

HTML Colors

```
<html>
<body>
<h1 style="border: 5px solid red;">UNIVERSITY</h1>
<h2 style="border: 5px solid Blue;"> UNIVERSITY </h2>
<h4 style="border: 9px solid green;"> UNIVERSITY </h4>
</body>
</html>
```

My First Heading

UNIVERSITY

UNIVERSITY

UNIVERSITY

HTML Colors (HSL and HSLA)

- **HSL** stands for **hue, saturation, and lightness**.
- **HSLA** color values are an extension of HSL with an **Alpha channel** (opacity).
 - Hue is a degree on the color wheel from 0 to 360. 0 is red, 120 is green, and 240 is blue.
 - Saturation is a percentage value. 0% means a shade of gray, and 100% is the full color.
 - Lightness is also a percentage value. 0% is black, and 100% is white.

HTML Colors (HSL and HSLA)

- **HSL** stands for **hue, saturation, and lightness**.
- Varying *hue* values
 - Hue is a degree on the color wheel from 0 to 360. **0 is red, 120 is green, and 240 is blue**

`hsl(0, 100%, 50%)`

`hsl(240, 100%, 50%)`

`hsl(147, 50%, 47%)`

`hsl(300, 76%, 72%)`

`hsl(39, 100%, 50%)`

`hsl(248, 53%, 58%)`

HTML Colors (HSL and HSLA)

- HSL stands for **hue, saturation, and lightness**.
- Varying ***saturation*** values
 - 100% is pure color, no shades of gray.
 - 50% is 50% gray, but you can still see the color.
 - 0% is completely gray; you can no longer see the color.

<code>hsl(0, 100%, 50%)</code>	<code>hsl(0, 80%, 50%)</code>
<code>hsl(0, 60%, 50%)</code>	<code>hsl(0, 40%, 50%)</code>
<code>hsl(0, 20%, 50%)</code>	<code>hsl(0, 0%, 50%)</code>

HTML Colors (HSL and HSLA)

- HSL stands for **hue, saturation, and lightness**.
- Varying *lightness* values
 - 0% means no light (black).
 - 50% means 50% light (neither dark nor light).
 - 100% means full lightness (white).

<code>hsl(0, 100%, 0%)</code>	<code>hsl(0, 100%, 25%)</code>
<code>hsl(0, 100%, 50%)</code>	<code>hsl(0, 100%, 75%)</code>
<code>hsl(0, 100%, 90%)</code>	<code>hsl(0, 100%, 100%)</code>

HTML Colors (HSL and HSLA)

- **HSLA** stands for **hue, saturation, lightness, and Alpha**.
- The alpha parameter is a number between **0.0 (fully transparent)** and **1.0 (not transparent at all)**



HTML Colors

```
<html>
<style>
div {
background-color: hsl(150, 50%, 60%);
color: hsl(100%, 0%, 0%);
padding: 20px;
}
</style>
<body>
<div>
<h1> India</h1>
<p>Around the globe, India is a representative of unity in diversity. India is a land of different cultures,
    castes, creeds, religions; despite the many differences we continue to live harmoniously. Indians are
    peace-loving and go out of the way to help people at times of peril. We believe in the motto, "Atithi
    Devo Bhava" which means our guests are our gods and are especially helpful and kind towards
    tourists visiting our nation.
</p>
</div>
</body>
</html>
```

India

Around the globe, India is a representative of unity in diversity. India is a land of different cultures, castes, creeds, religions; despite the many differences we continue to live harmoniously. Indians are peace-loving and go out of the way to help people at times of peril. We believe in the motto, "Atithi Devo Bhava" which means our guests are our gods and are especially helpful and kind towards tourists visiting our nation.

HTML BORDER Colors

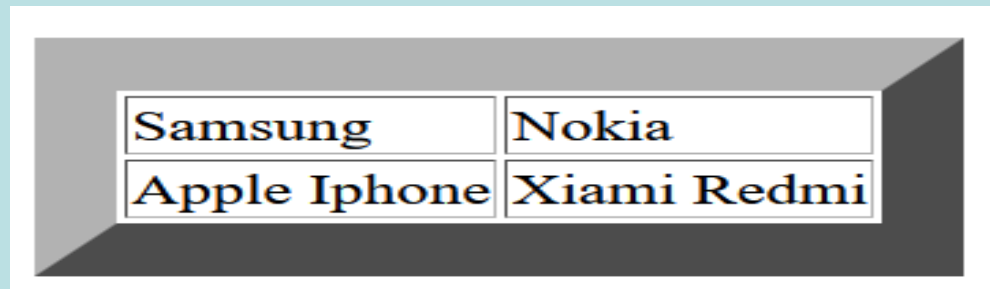
```
<html>
```

```
<head>
```

```
<TABLE BORDER=20 BORDERCOLORLIGHT= BLUE  
  BORDERCOLORDARK= ROSE> <TR> <TD> Samsung</TD>  
  <TD> Nokia</TD> </TR> <TR> <TD> Apple Iphone</TD> <TD>  
  Xiami Redmi</TD> </TR> </TABLE>
```

```
</head>
```

```
</html>
```



Samsung	Nokia
Apple Iphone	Xiami Redmi

HTML Images

```
<html>
  <head>
    <title>VIT Bhopal</title>
  </head>
  <body>
    <div class="container">
      <div class="image">
        
      </div>
      <div class="text">
        <h1> Academic Block.</h1>
      </div>
    </div>
  </body>
</html>
```



Academic Block.

HOW TO MAKE A LINK

1) The tags used to produce links are the `<a>` and `</a>`. The `<a>` tells where the link should start and the `</a>` indicates where the link ends. Everything between these two will work as a link.

```
<html>
```

```
<body>
```

```
<h1>HTML Links</h1>
```

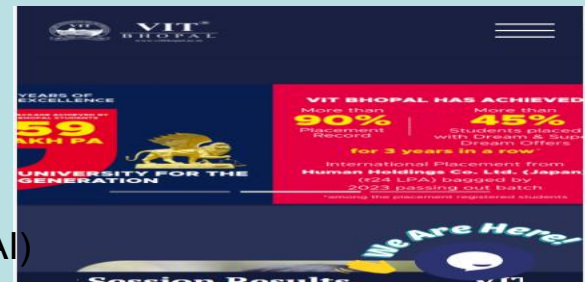
```
<p><a href=" https://vitbhopal.ac.in/ ">Visit VIT  
Bhopal!</a></p>
```

```
</body>
```

```
</html>
```

[Visit VIT Bhopal!](https://vitbhopal.ac.in/)

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Link to an Email Address

```
<html>
```

```
<body>
```

```
<h2>Link to an Email Address</h2>
```

```
<p>To create a link that opens in the user's email  
program (to let them send a new email), use mailto:  
inside the href attribute:</p>
```

```
<p><a  
href="mailto:adyashasahu@vitbhopal.ac.in">Send  
email</a></p>
```

```
</body>
```

```
</html>
```

Link to an Email Address

To create a link that opens in the user's email program (to let them send a new email), use mailto: inside the href attribute:

[Send email](mailto:adyashasahu@vitbhopal.ac.in)

Tables

The `<TABLE></TABLE>` element has four sub-elements:

Table Row `<TR></TR>`.

Table Header `<TH></TH>`.

Table Data `<TD></TD>`.

Caption `<CAPTION></CAPTION>`.

The table row elements usually contain table header elements or table data elements.

Tables

```
<table border="1">  
<tr>  
<th> Column 1 header </th>  
<th> Column 2 header </th>  
</tr>  
<tr>  
<td> Row1, Col1 </td>  
<td> Row1, Col2 </td>  
</tr>  
<tr>  
<td> Row2, Col1 </td>  
<td> Row2, Col2 </td>  
</tr>  
</table>
```

Column 1 header	Column 2 header
Row1, Col1	Row1, Col2
Row2, Col1	Row2, Col2

Tables Attributes

- **BGColor**: Some browsers support background colors in a table.
- **Width**: you can specify the table width as an absolute number of pixels or a percentage of the document width. You can set the width for the table cells as well.
- **Border**: You can choose a numerical value for the border width, which specifies the border in pixels.
- **CellSpacing**: Cell Spacing represents the space between cells and is specified in pixels.

Table Attributes

- **CellPadding**: Cell Padding is the space between the cell border and the cell contents and is specified in pixels.
- **Align**: tables can have left, right, or center alignment.
- **Background**: Background Image, will be titled in IE3.0 and above.
- **BorderColor, BorderColorDark**.

Table Caption

- A table caption allows you to specify a line of text that will appear centered above or below the table.

<TABLE BORDER=1 CELLPADDING=2>

**<CAPTION ALIGN="BOTTOM"> Label For My Table
</CAPTION>**

- The Caption element has one attribute ALIGN that can be either TOP (Above the table) or BOTTOM (below the table).

Table Header

- Table Data cells are represented by the TD element. Cells can also be TH (Table Header) elements which results in the contents of the table header cells appearing **centered and in bold text**.

Table Data and Table Header Attributes

- **Colspan:** Specifies how many cell columns of the table this cell should span.
- **Rowspan:** Specifies how many cell rows of the table this cell should span.
- **Align:** cell data can have left, right, or center alignment.
- **Valign:** cell data can have top, middle, or bottom alignment.
- **Width:** you can specify the width as an absolute number of pixels or a percentage of the document width.
- **Height:** You can specify the height as an absolute number of pixels or a percentage of the document height.

Basic Table Code

```
<TABLE BORDER=1 width=50%>
<CAPTION> <h1>Spare Parts </h1> </Caption>
<TR><TH>Stock Number</TH><TH>Description</TH><TH>List Price</TH></TR>
<TR><TD bgcolor=red>3476-AB</TD><TD>76mm
Socket</TD><TD>45.00</TD></TR>
<TR><TD>3478-AB</TD><TD><font color=blue>78mm Socket</font>
</TD><TD>47.50</TD></TR>
<TR><TD>3480-AB</TD><TD>80mm Socket</TD><TD>50.00</TD></TR>
</TABLE>
```

Spare Parts

Stock Number	Description	List Price
3476-AB	76mm Socket	45.00
3478-AB	78mm Socket	47.50
3480-AB	80mm Socket	50.00

Table Data and Table Header Attributes

`<Table border=1 cellpadding =2>`

`<tr> <th> Column 1 Header</th> <th>
Column 2 Header</th> </tr>`

`<tr> <td colspan=2> Row 1 Col 1</td> </tr>`

`<tr> <td rowspan=2>Row 2 Col 1</td>`

`<td> Row 2 Col2</td> </tr>`

`<tr> <td> Row 3 Col2</td> </tr>`

`</table>`

Column 1 Header	Column 2 Header
Row 1 Col 1	
Row 2 Col 1	Row 2 Col2
	Row 3 Col2

Exercise

- Create an HTML page with a **colored background** and **custom font color**.
- Insert at least **one image** related to chemistry (e.g., periodic table, lab equipment, molecules). **Add the source link** of the image used.
- Display the following chemical equations using proper formatting with **<sub>** and **<sup>** tags.
 - $\text{C}_2\text{H}_5\text{OH} + \text{PCl}_5 = \text{C}_2\text{H}_5\text{Cl} + \text{POCl}_3 + \text{HCl}$
 - $4\text{H}_3\text{PO}_3 = 3\text{H}_3\text{PO}_4 + \text{PH}_3$
 - $\text{PCl}_3 + \text{Cl}_2 = \text{PCl}_5$
- **Draw a border** around the section that includes the chemical equations.
- **Create a table** showing at least 5 elements with the following columns:
 - Element Name
 - Symbol
 - Atomic Number
 - State (Solid/Liquid/Gas)

Try using different border styles (dotted, dashed, solid). Add a heading using **<h1>** and a brief paragraph describing your page.

Exercise

- Write the HTML code to create a basic HTML5 document with a title “My First Web Page” and a heading that says “Hello, World!”.
- Write HTML to display the following line:
“HTML is Easy to Learn”
 - i. Make the word **HTML** bold
 - ii. Make the word **Easy** italic
 - iii. Underline the word **Learn**
- Create a webpage with a light blue background and display a paragraph in red text saying “This is a colored paragraph”.

LIST

Unordered HTML List

- An unordered list starts with the `<ul>` tag. Each list item starts with the `<li>` tag.
- The list items will be marked with bullets (small black circles) by default.

An unordered HTML list

- Coffee
- Tea
- Milk

LIST

Ordered HTML List

- An ordered list starts with the `<ol>` tag. Each list item starts with the `<li>` tag.
- The list items will be marked with numbers by default.

An ordered HTML list

1. Coffee
2. Tea
3. Milk

LIST

HTML Description Lists

- HTML also supports description lists.
- A description list is a list of terms, with a description of each term.
- The `<dl>` tag defines the description list, the `<dt>` tag defines the term (name), and the `<dd>` tag describes each term:

A Description List

Coffee

- black hot drink

Milk

- white cold drink

LIST

HTML List Tags

Tag	Description
<u></u>	Defines an unordered list
<u></u>	Defines an ordered list
<u></u>	Defines a list item
<u><dl></u>	Defines a description list
<u><dt></u>	Defines a term in a description list
<u><dd></u>	Describes the term in a description list

LIST

```
<head>
  <title>HTML Nested Unordered List</title>
</head>
<body>
  <ul>
    <li>Name</li>
    <ul>
      <li>Abhay</li>
      <li>Raghav</li>
      <li>Rishi</li>
      <li>Saket</li>
    </ul>
    <li>Subject</li>
    <ul>
      <li>DS</li>
      <li>DSA</li>
      <li>Python</li>
    </ul>
    <li>JAVA</li>
    <li>CN</li>
  </ul>
</body>
</html>
```

- Name
 - Abhay
 - Raghav
 - Rishi
 - Saket
- Subject
 - DS
 - DSA
 - Python
- JAVA
- CN

FORMS

- Forms add the ability to web pages to not only provide the person viewing the document with dynamic information but also to obtain information from the person viewing it, and process the information.

Objectives:

Upon completing this section, you should be able to

1. Create a FORM.
 2. Add elements to a FORM.
 3. Define CGI (**Common Gateway Interface**).
 4. Describe the purpose of a CGI Application.
 5. Specify an action for the FORM.
- Forms work in all browsers.
 - Forms are Platform Independent.

FORMS

- To insert a form we use the `<FORM></FORM>` tags. The rest of the form elements must be inserted in between the form tags.

```
<html>
```

```
<body>
```

```
<h1>The form action attribute</h1>
```

```
<form action="/action_page.php">
```

—————→ Form action -**Server**

```
<label for="fname">First name:</label>
```

```
<input type="text" id="fname" name="fname"><br><br>
```

```
<label for="lname">Last name:</label>
```

```
<input type="text" id="lname" name="lname"><br><br>
```

```
<input type="submit" value="Submit">
```

```
</form>
```

```
<p>Click the "Submit" button and the form-data will be sent to a page on the  
server called "action_page.php".</p>
```

```
</body>
```

```
</html>
```

FORMS

The form action attribute

First name:

Last name:

Submit

Click the "Submit" button and the form-data will be sent to a page on the server called "action_page.php".

<FORM> element attributes

- **ACTION**: is the **URL** of the **CGI** (Common Gateway Interface) program that is going to accept the data from the form, process it, and send a response back to the browser.
- **METHOD**: **GET** (default) or **POST** specifies which **HTTP** method will be used to send the form's contents to the web server. The CGI application should be written to accept the data from either method.
- **NAME**: is a form name used by **VBScript** or **JavaScripts**.
- **TARGET**: is the target frame where the response page will show up.

Form Elements

- Form elements have properties: **Text** boxes, **Password** boxes, **Checkboxes**, **Option(Radio)** buttons, **Submit**, **Reset**, **File**, **Hidden** and **Image**.
- The properties are specified in the **TYPE** Attribute of the HTML element **<INPUT></INPUT>**.

Name:

Sami Ali

Student No.

123456789

Address:

Al al-Bayt University
CIS Department
Faculty of IT

City:

Amman

Amman

Irbed

Karak

is foreign?



Male:



Female:



Submit

Reset

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Form Elements

<INPUT> Element's Properties

TYPE= Type of INPUT entry field.

NAME = Variable name passed to CGI application

VALUE= The data associated with the variable name to be passed to the CGI application

CHECKED= Button/box checked

SIZE= Number of visible characters in text field

MAXLENGTH= Maximum number of characters accepted.

Text Box

- **Text boxes**: Used to provide input fields for text, phone numbers, dates, etc.

<INPUT TYPE= " TEXT " >



Browser will display

Textboxes use the following attributes:

- **TYPE**: text.
- **SIZE**: determines the size of the textbox in characters. **Default=20** characters.
- **MAXLENGTH** : determines the maximum number of characters that the field will accept.
- **NAME**: is the name of the variable to be sent to the CGI application.
- **VALUE**: will display its contents as the default value.

Example on Text Box

```
<TITLE>Form_Text_Type</TITLE>
</HEAD> <BODY>
<h1> <font color=blue>Please enter the following
    bioData</font></h1>
<FORM name="fome1"  Method= " get " Action= " URL " >
First Name: <INPUT TYPE="TEXT" NAME="FName"
SIZE="15" MAXLENGTH="25"><BR>
Last Name: <INPUT TYPE="TEXT" NAME="LName"
SIZE="15" MAXLENGTH="25"><BR>
Nationality: <INPUT TYPE="TEXT" NAME="Country"
SIZE="25" MAXLENGTH="25"><BR>
The Phone Number: <INPUT TYPE="TEXT" NAME="Phone"
SIZE="15" MAXLENGTH="12"><BR>
</FORM> </BODY> </HTML>
```

Example on Text Box

Method	Description
get	Appends form data to the URL (visible in the address bar). Used for searches or data retrieval .
post	Sends form data inside the request body (not visible in the URL). Used for submitting sensitive or large amounts of data .

Example on Text Box

Please enter the following bioData

First Name:

Last Name:

Nationality:

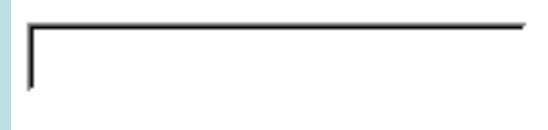
The Phone Number:

Password

- **Password:** Used to allow entry of passwords.

<INPUT TYPE= " PASSWORD " >

Browser will display



Text typed in a password box is starred out in the browser display.

Password boxes use the following attributes:

- **TYPE:** password.
- **SIZE:** determines the size of the textbox in characters.
- **MAXLENGTH:** determines the maximum size of the password in characters.
- **NAME:** is the name of the variable to be sent to the CGI application.
- **VALUE:** is usually blank.

Hidden

- **Hidden:** Used to send data to the CGI application that you don't want the web surfer to see, change or have to enter but is necessary for the application to process the form correctly.

<INPUT TYPE="HIDDEN">

Nothing is displayed in the browser.

Hidden inputs have the following attributes:

- **TYPE:** hidden.
- **NAME:** is the name of the variable to be sent to the CGI application.
- **VALUE:** is usually set a value expected by the CGI application.

Check Box

- **Check Box:** Check boxes allow the users to select more than one option.

<INPUT TYPE="CHECKBOX">

Browser will display



Checkboxes have the following attributes:

- **TYPE:** checkbox.
- **CHECKED:** is blank or CHECKED as the initial status.
- **NAME:** is the name of the variable to be sent to the CGI application.
- **VALUE:** is usually set to a value.

```
<HTML> <HEAD><TITLE>CheckBoxType</TITLE> </HEAD>
<BODY>
<h1> <font color=green>Please check one of the
following</font></h1>
<FORM name="fome3" Action="url" method="get">
<font color=red> Select Country: </font><BR>
jordan:<INPUT TYPE="CheckBox" Name="country"
CHECKED><BR>
Yemen<INPUT TYPE="CheckBox" Name="country"><BR>
Qatar:<INPUT TYPE="CheckBox" Name="country"><BR>
<BR>
<font color=blue>Select Language:</font><BR>
Arabic:<INPUT TYPE="CheckBox" Name="language"
CHECKED><BR> English:<INPUT TYPE="CheckBox"
Name="language"><BR>
French:<INPUT TYPE="CheckBox" Name="language">
<BR></FORM> </BODY></HTML>
```

Select Country:

jordan: ☒

Yemen ☐

Qatar: ☐

Select Language:

Arabic: ☒

English: ☐

French: ☐

Radio Button

- **Radio Button:** Radio buttons allow the users to select only one option.

<INPUT TYPE="RADIO">

Browser will display



Radio buttons have the following attributes:

- **TYPE:** radio.
- **CHECKED:** is blank or CHECKED as the initial status. Only one radio button can be checked
- **NAME:** is the name of the variable to be sent to the CGI application.
- **VALUE:** usually has a set value.

```
<HTML> <HEAD><TITLE>CheckBoxType</TITLE> </HEAD>
<BODY>
<h1> <font color=green>Please check one of the
following</font></h1>
<FORM name="fome3" Action="url" method="get">
<font color=red> Select Country: </font><BR>
jordan:<INPUT TYPE="RADIO" Name="country"
CHECKED><BR>
Yemen<INPUT TYPE="RADIO " Name="country"><BR>
Qatar:<INPUT TYPE="RADIO" Name="country"><BR>
<BR>
<font color=blue>Select Language:</font><BR>
Arabic:<INPUT TYPE="RADIO" Name="language"
CHECKED><BR> English:<INPUT TYPE=" RADIO "
Name="language"><BR>
French:<INPUT TYPE=" RADIO " Name="language">
<BR></FORM> </BODY></HTML>
```

Please check one of the following

Select Country:

jordan: ☒

Yemen:

Qatar: ☐

Select Language:

Arabic: ☒

English:

French:

```
<HTML><HEAD>  
<TITLE>RADIOBox</TITLE> </HEAD>  
<BODY>
```

Form #1:

```
<FORM>  
  <INPUT TYPE="radio" NAME="choice" VALUE="one"> Yes.  
  <INPUT TYPE="radio" NAME="choice" VALUE="two"> No.  
</FORM>  
<HR color=red size="10" >
```

Form #2:

```
<FORM>  
  <INPUT TYPE="radio" NAME="choice" VALUE="three"  
CHECKED> Yes.  
  <INPUT TYPE="radio" NAME="choice" VALUE="four"> No.  
</FORM>  
</BODY></HTML>
```

Form #1:

☐ Yes. ☒ No.

Form #2:

☒ Yes. ☐ No.

Push Button

- **Push Button:** This element would be used with JavaScript to cause an action to take place.

<INPUT TYPE="BUTTON">

Browser will display



Push Button has the following attributes:

- **TYPE:** button.
- **NAME:** is the name of the button to be used in scripting.
- **VALUE:** determines the text label on the button.

**<DIV align=center>

**

<FORM>

**<h1>Press Here to see a baby crying:
**

**<INPUT TYPE="button"
VALUE="PressMe">

**

**Click Here to see a baby shouting:
**

**<INPUT TYPE="button" VALUE="ClickMe" >

**

**Hit Here to see a baby eating:
**

**<INPUT TYPE="button" VALUE="HitME" >

**

</FORM></DIV>

Press Here to see a baby crying:

PressMe

Click Here to see a baby shouting:

ClickMe

Hit Here to see a baby eating:

HitME

Submit Button

- **Submit:** Every set of Form tags requires a Submit button. This is the element causes the browser to send the names and values of the other elements to the CGI Application specified by the ACTION attribute of the FORM element.

<INPUT TYPE="SUBMIT">

The browser will display



Submit has the following attributes:

- **TYPE:** submit.
- **NAME:** value used by the CGI script for processing.
- **VALUE:** determines the text label on the button, usually Submit Query.

```
<FORM Action="URL" method="get">  
First Name: <INPUT TYPE="TEXT" Size=25  
name="firstName"><BR>  
Family Name: <INPUT TYPE="TEXT" Size=25  
name="LastName"><BR>  
<BR>  
<FONT Color=red>  
Press Here to submit the data:<BR>  
<INPUT TYPE="submit" VALUE="SubmitData " >  
</FORM>
```

First Name:
Family Name:

Press Here to submit the data:

Dr. Adyasha Sahu (SCAI)

Reset Button

- **Reset:** It is a good idea to include one of these for each form where users are entering data. It allows the surfer to clear all the input in the form.
- **<INPUT TYPE="RESET">**
- Browser will display 
-
- Reset buttons have the following attributes:
- **TYPE:** reset.
- **VALUE:** determines the text label on the button, usually Reset.

```
<FORM Action="URL" method="get">
First Name: <INPUT TYPE="TEXT" Size=25
name="firstName"> <BR>
Family Name: <INPUT TYPE="TEXT" Size=25
name="LastName"><BR>
<BR>
<FONT Color = red>
<STRONG><font size=5>Press Here to submit
the data:</font></STRONG><BR>
<INPUT TYPE="submit" VALUE="SubmitData">
<INPUT TYPE="RESET" VALUE="Reset">
</FORM>
```

First Name:

Family Name:

Press Here to submit the data:

Dr. Advasha Sahu (SCAI)

Other Elements used in Forms

- The two following examples are **<SELECT></SELECT>** elements, where the attributes are set differently.

The Select elements attributes are:

- **NAME**: is the name of the variable to be sent to the CGI application.
- **SIZE**: this sets the number of **visible** choices.
- **MULTIPLE**: the presence of this attribute signifies that the user can make multiple selections. By default only one selection is allowed.

Other Elements used in Forms

- **List Box:**



- **Name:** is the name of the variable to be sent to the CGI application.
- **SIZE:** is greater than one.

Other Elements used in Forms

- **Option**

The list items are added to the **<SELECT>** element by inserting **<OPTION></OPTION>** elements.

The Option Element's attributes are:

- **SELECTED**: When this attribute is present, the option is selected when the document is initially loaded. **It is an error for more than one option to be selected.**
- **VALUE**: Specifies the value the variable named in the select element.

</HEAD>

<BODY>

<h2>What type of Computer do you have?<h2>

<FORM>

<SELECT NAME="ComputerType" size=4>

<OPTION value="IBM" SELECTED> IBM</OPTION>

<OPTION value="INTEL"> INTEL</OPTION>

<OPTION value=" Apple"> Apple</OPTION>

<OPTION value="Compaq"> Compaq</OPTION>

</SELECT>

</FORM></BODY></HTML>

What type of Computer do you have?

IBM
INTEL
Apple
Compaq

There are eleven different types of form elements:

Button

Checkbox

☐

FileUpload

Hidden

Password

Radio

☐

Reset object

Select object

Submit object

Text

Textarea

Text area

`<label for="story">Tell us your story:`

`</label>`

`<textarea id="story" name="story"`

`rows="5" cols="33">`

"I use goal-setting regularly in my career and life. I try to make my goals clear and specific with deadlines....`</textarea>`

Tell us your story:

"I use goal-setting regularly in my career and life. I try to make my goals clear and specific with deadlines....

Exercise

Write the HTML code to create a **College Event Registration Form** that collects the following information:
(method="post", action="register_event.php")

1. Personal Details (Full Name, Student ID, school, programme, year, Email Address, Mob no, Gender)

2. Event Choices

- Event Type (drop-down <select> with options; default "Select an event")
- Number of Guests
- **T-Shirt Size** (Drop-down)

3. Preferences & Extras

- Meal Preference (Vegetarian, Non-Vegetarian, Vegan; required)
- Would you like a campus tour? (checkbox)
- Preferred Mode of Contact

4. Additional Info

- Arrival Date (date input, required)
- Special Requests (textarea with placeholder "Let us know any special requirements")
- How did you hear about us?
- Comment and Query Box

5. Form Controls

- A checkbox to agree to "College Event Terms & Conditions" (required)
- Submit button labeled "Register"
- Reset button to clear the form

6. Add a title for this form.

7. Use place holders

Exercise

Write the HTML code to create a **College Event Registration Form** that collects the following information:

1. Personal Details (Full Name, Student ID, Email Address)

2. Event Choices

- **Event Type** (drop-down <select> with options: Workshop, Seminar, Sports Meet, Cultural Fest; default “Select an event”)
- **Number of Guests**

3. Preferences & Extras

- Meal Preference (Vegetarian, Non-Vegetarian, Vegan; required)
- Would you like a campus tour? (checkbox)

4. Additional Info

- Arrival Date (date input, required)
- Special Requests (textarea with placeholder “Let us know any special requirements”)

5. Form Controls

- A checkbox to agree to “College Event Terms & Conditions” (required)
- Submit button labeled “Register”
- Reset button to clear the form

CANVAS

- The HTML `<canvas>` element is used to draw graphics on a web page.

```
<html>
```

```
<body>
```

```
<canvas id="myCanvas" width="200" height="100" style="border:1px solid #d3d3d3;">
```

```
Your browser does not support the HTML canvas tag.</canvas>
```

```
<script>
```

```
var c = document.getElementById("myCanvas");
```

```
var ctx = c.getContext("2d");
```

```
ctx.moveTo(0,0);
```

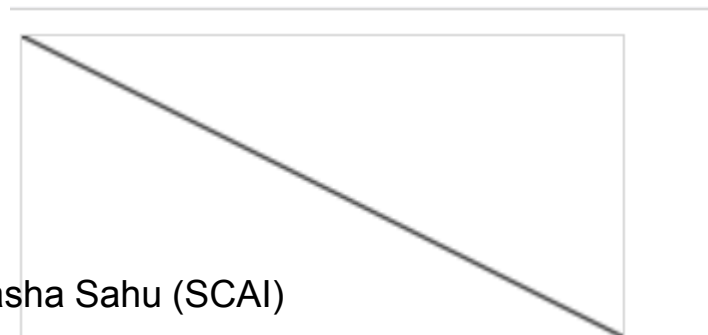
```
ctx.lineTo(200,100);
```

```
ctx.stroke();
```

```
</script>
```

```
</body>
```

```
</html>
```



MEDIA

```
<html>
```

```
<body>
```

```
<h1>The area media attribute</h1>
```

```
<p>Click on the sun to watch it closer:</p>
```

```

```

```
<map name="planetmap">
```

```
  <area shape="rect" coords="0,0,82,126" alt="Sun" href="https://hips.hearstapps.com/hmg-prod/images/gettyimages-1354937085641c7cd7541be.jpg?crop=0.668xw:1.00xh;0.167xw,0&resize=1200:*" media="screen and (min-color-index:256)">
```

```
</map>
```

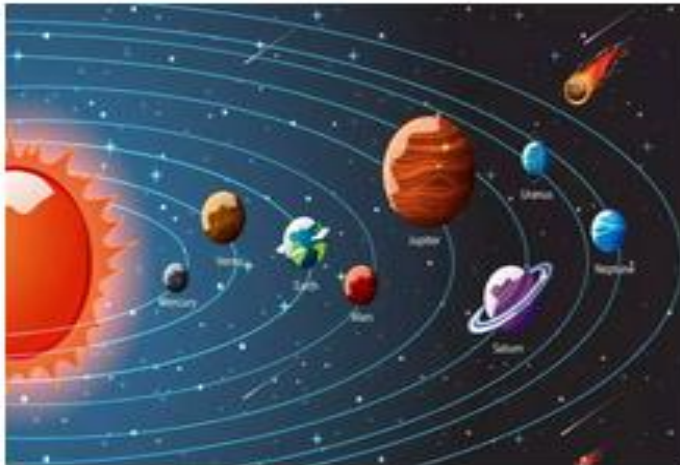
```
</body>
```

```
</html>
```

MEDIA

The area media attribute

Click on the sun to watch it closer:



ADVANTAGES OF HTML

- Easy to use
- Loose syntax (although, being too flexible will not comply with standards).
- Supported on almost every browser, if not all browsers.
- Widely used; established on almost every website, if not all websites.
- Very similar to XML syntax, which is increasingly used for data storage.
- Free - You need not buy any software.
- Easy to learn & code even for novice programmers.

DISADVANTAGES OF HTML

- It cannot produce dynamic output alone, since it is a static language
- Sometimes, the structuring of HTML documents is hard to grasp
- You have to keep up with deprecated tags, and make sure not to use them
- Deprecated tags appear because another language that works with HTML has replaced the original work of the tag; thus the other language needs to be learned (most of the time, it is CSS)
- Security features offered by HTML are limited